Simplification of $N_{2a} + N_{2b\,\text{finite}}$

Notation

$$a = x' - w', \qquad r = x - z, \qquad u = y - x, \qquad n = y - z, \qquad \xi = \frac{p^+}{k^+}. \tag{1}$$

These satisfy

$$n = u + r, \qquad n^2 - u^2 = 2\,n\cdot r - r^2, \qquad u\cdot r = n\cdot r - r^2, \qquad u\cdot n = n^2 - n\cdot r. \tag{2}$$

The two colour structures are

$$\mathcal{C}_2 = \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[- f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_c U^{af}(y)\Big]\rho^{b'}(x')\rho^{f}(y), \tag{3}$$

$$\mathcal{C}_z = \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c U^{af}(y) - N_c U^{af}(z)\Big]\rho^{b'}(x')\rho^{f}(y). \tag{4}$$

Every tensor bracket is contracted with $T^{im} = \frac{a^i(z-x)^m}{a^2r^2} = -\frac{a^i r^m}{a^2r^2}$; below we multiply by a^2r^2. Note that $a^2r^2\,\delta^{im}T^{im} = -a\cdot r$. In the finite term $d_\perp = 2$.

Step 1: contract each bracket

$$d_\perp \text{ term:} \quad \xi(1-\xi)\,\frac{u^2-n^2}{r^2}\,(-a\cdot r) = \xi(1-\xi)\,\frac{(a\cdot r)(2\,n\cdot r - r^2)}{r^2}, \tag{5}$$

$$(1-\xi)\delta \text{ term:} \quad \Big[\frac{n\cdot r}{r^2} + \frac{u\cdot n}{2u^2} - \frac{u^2-n^2}{2r^2}\Big] = \frac{2\,n\cdot r}{r^2} - \frac{1}{2} + \frac{u\cdot n}{2u^2} \;\Rightarrow\; -(1-\xi)(a\cdot r)\Big[\frac{2\,n\cdot r}{r^2} - \frac{1}{2} + \frac{u\cdot n}{2u^2}\Big], \tag{6}$$

$$\xi\delta \text{ term:} \quad \Big[\frac{u\cdot r}{r^2} - \frac{u\cdot n}{2n^2} - \frac{u^2-n^2}{2r^2}\Big] = \frac{2\,n\cdot r}{r^2} - \frac{3}{2} - \frac{u\cdot n}{2n^2} \;\Rightarrow\; -\xi(a\cdot r)\Big[\frac{2\,n\cdot r}{r^2} - \frac{3}{2} - \frac{u\cdot n}{2n^2}\Big], \tag{7}$$

$$\text{last bracket:} \quad (a\cdot r)\Big[\frac{2\,n\cdot r}{r^2} - 1\Big] - \frac{(a\cdot n)(u\cdot r)}{2n^2} + \frac{(a\cdot u)(n\cdot r)}{2u^2}, \tag{8}$$

$$+B \text{ term:} \quad a^2r^2\,T^{im}B^{im} = -(a\cdot n) - \frac{(a\cdot n)(u\cdot r)}{2u^2}, \tag{9}$$

$$[k^+/p^+]_+B \text{ term:} \quad -\Big[\frac{k^+}{p^+}\Big]_+ a^2r^2\,T^{im}B^{im} = +\Big[\frac{k^+}{p^+}\Big]_+ G_p, \qquad G_p = (a\cdot n)\Big[1 + \frac{u\cdot r}{2u^2}\Big], \tag{10}$$

$$q^+ \text{ block:} \quad a^2r^2\,T^{im}A^{im} = -G_q, \qquad G_q = (a\cdot u)\Big[1 - \frac{n\cdot r}{2n^2}\Big]. \tag{11}$$

Step 2: collect by powers of ξ

The $\frac{2n\cdot r}{r^2}$ pieces of the $(1-\xi)\delta$, $\xi\delta$ and last brackets cancel, because $-(1-\xi) - \xi + 1 = 0$. The constants give $(a\cdot r)\big[\frac{1-\xi}{2} + \frac{3\xi}{2} - 1\big] = \xi(a\cdot r) - \frac{1}{2}(a\cdot r)$. The regular part of the kernel is therefore

$$\mathcal{R}(\xi) = \xi(1-\xi)\,\mathcal{A} + \xi\,\mathcal{B} + \mathcal{C}, \tag{12}$$

with

$$\mathcal{A} = \frac{(a\cdot r)\,(2\,n\cdot r - r^2)}{r^2}, \tag{13}$$

$$\mathcal{B} = (a\cdot r)\Big[1 + \frac{u\cdot n}{2u^2} + \frac{u\cdot n}{2n^2}\Big], \tag{14}$$

$$\mathcal{C} = -\frac{a\cdot r}{2} - a\cdot n - \frac{(a\cdot r)(u\cdot n) + (a\cdot n)(u\cdot r) - (a\cdot u)(n\cdot r)}{2u^2} - \frac{(a\cdot n)(u\cdot r)}{2n^2}. \tag{15}$$

Step 3: simplify $\mathcal{C}$

Use $a\cdot u = a\cdot n - a\cdot r$, $u\cdot n = n^2 - n\cdot r$ and $u\cdot r = n\cdot r - r^2$:

$$\begin{aligned}
&(a\cdot r)(u\cdot n) + (a\cdot n)(u\cdot r) - (a\cdot u)(n\cdot r)\\
&= (a\cdot r)n^2 - (a\cdot r)(n\cdot r) + (a\cdot n)(n\cdot r) - (a\cdot n)r^2 - (a\cdot n)(n\cdot r) + (a\cdot r)(n\cdot r)\\
&= (a\cdot r)\,n^2 - (a\cdot n)\,r^2,
\end{aligned} \tag{16}$$

so that

$$\mathcal{C} = -\frac{a\cdot r}{2} - a\cdot n - \frac{(a\cdot r)\,n^2 - (a\cdot n)\,r^2}{2u^2} - \frac{(a\cdot n)(u\cdot r)}{2n^2}. \tag{17}$$

Steps 1–3 were checked numerically at random points: $\xi(1-\xi)\mathcal{A} + \xi\mathcal{B} + \mathcal{C}$ equals the original contracted bracket (including the $+B$ term) to 10^{-15}.

Step 4: the log terms

$$-\frac{T^{im}B^{im}}{(y-z)^2} = \frac{G_p}{a^2r^2n^2}, \qquad -\frac{T^{im}A^{im}}{(y-x)^2} = \frac{G_q}{a^2r^2u^2}, \qquad K_S = \frac{(a\cdot n)(u\cdot n)}{a^2\,n^2\,u^2\,r^2}. \tag{18}$$

Result

In the original variables:

$$\begin{aligned}
\mathcal{A} &= \frac{(x'-w')\cdot(x-z)\,\left[2\,(y-z)\cdot(x-z) - (x-z)^2\right]}{(x-z)^2},\\
\mathcal{B} &= (x'-w')\cdot(x-z)\left[1 + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} + \frac{(y-x)\cdot(y-z)}{2(y-z)^2}\right],\\
\mathcal{C} &= -\frac{(x'-w')\cdot(x-z)}{2} - (x'-w')\cdot(y-z) - \frac{(x'-w')\cdot(x-z)\,(y-z)^2 - (x'-w')\cdot(y-z)\,(x-z)^2}{2(y-x)^2}\\
&\quad - \frac{(x'-w')\cdot(y-z)\,(y-x)\cdot(x-z)}{2(y-z)^2},\\
G_p &= (x'-w')\cdot(y-z)\left[1 + \frac{(y-x)\cdot(x-z)}{2(y-x)^2}\right], \qquad G_q = (x'-w')\cdot(y-x)\left[1 - \frac{(y-z)\cdot(x-z)}{2(y-z)^2}\right],
\end{aligned} \tag{19}$$

and

$$\begin{aligned}
&\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} + \frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{\\
&\frac{e^{-ik\cdot(z-x)\frac{p^+}{k^+}}}{\left[p^+(y-x)^2 + (k^+-p^+)(y-z)^2\right](x'-w')^2(x-z)^2}\left[\frac{p^+(k^+-p^+)}{k^{+2}}\,\mathcal{A} + \frac{p^+}{k^+}\,\mathcal{B} + \mathcal{C} + \left[\frac{k^+}{p^+}\right]_+ G_p\right]\\
&\quad\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(y)\Big]\rho^{b'}(x)\,\rho^f(y)\\
&\quad-\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}} + \left[\frac{k^+}{k^+-p^+}\right]_+ + \left[\frac{k^+}{p^+}\right]_+ - 2\right]\frac{(x'-w')\cdot(y-z)\,(y-x)\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(y-x)^2\,(x-z)^2}\\
&\quad\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y) - N_c\,U^{af}(z)\Big]\rho^{b'}(x)\,\rho^f(y)\Bigg\}\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'}\frac{e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}}{\left[(k^+-q^+)(y-x)^2 + q^+(y-z)^2\right](x'-w')^2(x-z)^2}\,G_q\left\{\left[\frac{k^+}{q^+}\right]_+ - 1\right\}\\
&\quad\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(y)\Big]\rho^{b'}(x)\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'}\left\{\left[\frac{e^{-ik\cdot(w'-z)}\,G_p}{(x'-w')^2(x-z)^2(y-z)^2} + \frac{e^{-ik\cdot(w'-x)}\,G_q}{(x'-w')^2(x-z)^2(y-x)^2}\right]\right.\\
&\quad\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(y)\Big]\rho^{b'}(x)\,\rho^f(y)
\end{aligned}$$

$$- 2\, e^{-ik\cdot(w'-z)}\, \frac{(x'-w')\cdot(y-z)\,(y-x)\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(y-x)^2\,(x-z)^2} \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\}. \tag{20}$$

Notes

- The q^+ integral has been turned around: $\int_{k^+}^{0} dq^+\, T^{im} A^{im}(\cdots) = +\int_{0}^{k^+} dq^+\, \frac{G_q}{a^2 r^2}(\cdots)$.
- The two log signs flip to + because $-T^{im}B^{im} = +G_p/(a^2r^2)$ and $-T^{im}A^{im} = +G_q/(a^2r^2)$.
- The subtraction log keeps $-2 \times \frac{g^4}{16\pi^5} = -\frac{g^4}{8\pi^5}$.
- The full N_2 is this result plus $N_{2b\,\mathrm{UV}}$, which is unchanged.